

Rui-Ning Zhang (Raine)

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<https://scholar.google.com/citations?user=GPA06N8AAAAJ&hl=en&oi=a>

Research Interests

Energy transition; Policy design; Environmental & Energy Economics

Education Background

University of Cambridge <i>Visiting Ph.D. student, Energy Policy Research Group, Judge business school</i>	Cambridge, UK <i>09/2025-09/2026 (Expected)</i>
Beijing Institute of Technology (BIT) <i>Ph.D. student, Energy Economics and Climate Economics</i>	Beijing, China <i>09/2023-2027 (Expected)</i>
Beijing Institute of Technology (BIT) <i>M.S. in Energy Economics and Climate Economics</i>	Beijing, China <i>09/2021-06/2023</i>
Southwestern University of Finance and Economics (SWUFE) <i>B.S. in Mathematics and Applied Mathematics; B. Econ. in Economics (Double Degree)</i>	Chengdu, China <i>09/2017-06/2021</i>

Publications

Zhang, R., Li, H., Zhou, Y., Cao, Y., & Ai, X. Fiscal support design for residential energy transition: A central-local government perspective. *Energy Economics*, 2026, 109326. (SCI, SSCI; Q1; IF=14.2)

Zhang RN, Li H, Zhong S, Li Y, Guo YY. Who spends more on energy? Examining energy inequality among elderly households. *Energy for Sustainable Development*, 2025, 89: 101859. (SCI; Q2; IF=4.9)

Zhang RN, Li WL, Li YH, Li H. Job losses or gains? The impact of supply-side energy transition on employment in China. *Energy*, 2024, 308:132804. (SCI; Q1; IF=8.9)

Zhang RN, Li H, Chen TQ, Hou BD. How does natural gas consumption affect human health? Empirical evidence from China. *Journal of Cleaner Production*. 2021, 320:128795. (SCI; Q1; IF=11.072)

Zhang RN, Ai XN, Li H. How to design subsidy policies for clean energy projects? A study on “coal-to-gas” project in China. *Resources Policy*, 2023, 85: 103928. (SSCI; JCR Q1)

Li H, **Zhang RN**, Ai XN. Cost estimation of “coal-to-gas” project: Government and residents’ perspectives. *Energy Policy*. 2022, 167: 113077. (SSCI, SCI; Q1; IF=9)

Li H, **Zhang RN**, Guo YY. On-site and embodied water demand of urban residential buildings in China under Shared Socioeconomic Pathways. Review and Resubmitted to *Environmental Impact Assessment Review* (SSCI; JCR Q1; IF=11.2). April, 2026.

Li H, Zhao J, **Zhang RN**, Hou BD. The natural gas consumption and mortality nexus: A mediation analysis. *Energy*, 2022, 248: 123577. (SCI; Q1; IF=8.9)

Li H, Ai XN, Wang LL, **Zhang RN**. Substitution strategies for cooking energy: To use gas or electricity? *Journal of Environmental Management*, 2021, 303: 114135. (SCI; Q1; IF=8.7)

Zhou Z, Huang JP, Wang XK, **Zhang RN**, Li H, Zhao CG. Emission-mitigation-driven Export tax rebate policy: An evolutionary game analysis. *The Singapore Economic Review*, 2023, 1-22. (SSCI; Q2; IF=1.8)

Li H, **Zhang RN**, Zhang LY, Qin Y, Liao H. Energy production revolution in China during 2015–2019: progress and challenges. *Journal of Chinese Economic and Business Studies*, 2023, 21(1): 85-109. (ESCI)

Li H, **Zhang RN**. China Country Report in Energy Outlook and Energy-Saving Potential in East Asia 2023 (Kimura S, Phoumin H, and Purwanto AJ). Jakarta: ***ERIA***, 2023, 121-134. (*Non-Peer Reviewed Publications*)

Li H, **Zhang RN**. China Country Report in Analysis on Energy Cost of LCET-CN based on ERIA Energy Outlook Models 2024 (Kimura S, Setyawati C). Jakarta: ***ERIA***, 2024, 40-56. (*Non-Peer Reviewed Publications*)

Zhang RN, Li H. Opportunities, challenges and suggestions for China-ASEAN energy cooperation. ***Energy of China***, 2022, 44 (11): 77-86 (*Chinese Peer Reviewed Journal Article*). **张睿宁**, 李慧. 中国-东盟能源合作的机遇、挑战与对策建议. **中国能源**, 2022, 44(11): 77-86.

Li H, **Zhang RN**, Xu JH. Challenges and countermeasures of “coal-to-gas” project in China. ***International Petroleum Economics***, 2021, 29(10):35-41 (*Chinese Peer Reviewed Journal Article*). 李慧, **张睿宁**, 徐金红. 中国“煤改气”面临的挑战及对策建议. **国际石油经济**, 2021, 29(10): 35-41.

Li H, **Zhang RN**, Ai XN. Coal economic research current status, challenges and countermeasures for the high-quality development of coal industry in China. ***Coal Economic Research***. 2022,42(07):48-54 (*Chinese Peer Reviewed Journal Article*). 李慧, **张睿宁**, 艾先能. 我国煤炭行业高质量发展现状、问题与对策. **煤炭经济研究**, 2022, 42 (07): 48-54.

Li H, Li WL, Hu YM, **Zhang RN**, Li Yue, Chang YY. Analysis and trend prediction of refined oil price in 2024. ***Journal of Beijing Institute of Technology(Social Sciences Edition)***, 2024,26(02):59-67 (*Chinese Peer Reviewed Journal Article*). 李慧, 李威龙, 胡一鸣, **张睿宁**, 李玥, 常耀尹. 2024 年成品油价格分析与趋势预测. **北京理工大学学报(社会科学版)**,2024,26(02):59-67.

Working Papers

Zhang RN, Zhong S, Lee WJ, Li H. Population ageing and household energy affordability: Evidence from elderly empty-nest households in China.

Zhang RN, Hui Li, ReinerDM Liao H, Accounting for appliance heterogeneity unlocks greater mitigation benefits from residential efficiency upgrading.

Policy Consultation

Current Issues and Responses in the Transition of Rural Household Fuels, Policy Recommendations authored by Li H, **Zhang RN**, and Li Y, adopted by the **National Energy Administration of China**, November 2022.

Green and Low-Carbon Advanced Technology Demonstration Project, **National Energy Conservation Center (NECC)**, project organization contributor; received a letter of appreciation, 2024

Energy Economy Forecasting and Outlook Report: Analysis and Trend Forecast of Refined Oil Prices, contributor; reported by **CCTV News**, 2024

Peer Review Activities

Served as anonymous reviewer for ***Energy*** (JCR Q1; IF=8.9) since Jun 2025; for ***Energy Policy*** (JCR Q1; IF=9.2) since Nov 2025; for ***Cities*** (JCR Q1; IF=6.6) since Dec 2025; for ***Energy Research & Social Science*** (JCR Q1, IF=7.4) since April 2026.

Working experience

Teaching Assistant, *Energy Economics* (postgraduate course), Beijing Institute of Technology 09/2023-06/2024
Research Assistant, Cambridge Experimental & Behavioral Economics Group, University of Cambridge 10/2025

Software Copyrights

Residential Appliance Replacement Energy-Saving Analysis System V1.0 (Registration No. 2025SR2342527); Copyright holders: Li H, **Zhang RN**, Ji ZH, Li WL, Zheng YL; Registered in 2025

Residential "Coal-to-Gas" Techno-Economic Cost Analysis System V1.0 (Registration No. 2025SR0603750); Copyright holders: Li H, Li Y, **Zhang RN**, Zhou Y, Li WL; Registered in April, 2025

Cellulosic Ethanol Full-Chain Production Cost Analysis System V1.0 (Registration No. 2025SR0479576); Copyright holders: Li H, Sun Y, Mou WY, Li Y, **Zhang RN**; Registered in March, 2025.

Residents' Energy Substitution Burden Analysis System V1.0 (Registration No. 2024SR1904947); Copyright holders: Li H, **Zhang RN**, Sun Y, Li Y, Liao H; Registered in November, 2024.

Research Projects

National Natural Science Foundation of China (NSFC), General Program, 72473008, *Decision Mechanism, Impact Assessment, and Policy Design for Household Energy Clean Transition*, Approved in 2024, Participant.

National Natural Science Foundation of China (NSFC), Young Scientists Fund, 71804011, *Cost-Benefit Evaluation and Policy Research on 'Coal-to-Gas' in the Residential Sector*, 2019-01-01 to 2021-12-31, Participant.

Ministry of Science and Technology of the People's Republic of China, National Key R&D Program, 2022YFC2106304, *Environmental and Socioeconomic Impact Assessment of Bioenergy Refining*, 2023-03 to 2026-02, Participant.

Ministry of Education, Humanities and Social Sciences Research, Young Scientists Fund, *Cost-Benefit Evaluation and Policy Research on Household Energy Transition*, Approved in 2024, Participant.

Beijing Federation of Social Science Circles & Beijing Office of Philosophy and Social Sciences, Beijing Social Science Fund, 23JCC108, *Energy Substitution and Pathway Planning in Beijing Under the 'Dual Carbon' Goals*, 2024-01 to 2024-12, Participant.

Central Energy Enterprise Project, *Development and Support for an Oil and Gas Full-Chain Optimization and Decision-Making Model*, 2022-08 to 2023-08, Participant.

Ministry of Education, National Innovation and Entrepreneurship Training Program, *Research on Delivery Insurance Market and Pricing Models Under the Internet Economy*, 2019-04 to 2020-04, Project Leader.

Honors and Prizes

China National Scholarship issued by the Ministry of Education of the People’s Republic of China (Top 2%)	12/2024
China National Scholarship issued by the Ministry of Education of the People’s Republic of China (Top 2%)	12/2022
China National Scholarship issued by the Ministry of Education of the People’s Republic of China (Top 2%)	12/2021
Outstanding Graduate of Beijing, Beijing Municipal Education Commission (Top 2%)	06/2023
Outstanding Master's Thesis Award, Beijing Institute of Technology	06/2023
First Prize, Market Research and Analysis Competition (Beijing), China Business Statistics Society (Team Leader)	04/2024

Presentations

Young Energy Economists and Engineers Seminar (Florence, Italy) <i>Oral Presentation; Junior Reviewer</i>	04/2026
Winter Seminar: Developments in Global Energy Systems (Cambridge, UK) <i>On-site Operations Coordinator</i>	12/2025
China Academic Annual Conference of the Chinese Economic Association in Europe / Chinese Economic Association (UK) (Beijing, China) <i>Excellent Young Scholar Paper Award</i>	04/2025
Future Earth: Science and Applications Conference (Jingdezhen, Jiangxi, China) <i>Oral Presentation</i>	04/2025
Energy Economics and Management Annual Conference (Nanning, Guangxi, China) <i>Oral Presentation</i>	10/2024
Carbon Emissions and Sustainable Consumption Seminar (Chengdu, Sichuan, China) <i>Excellent Paper Award</i>	11/2023
Energy Economics and Management Annual Conference (Yan’an, Shaanxi, China) <i>Oral Presentation</i>	05/2023
The 6th UN-China Youth Environment Forum (Online) <i>Oral Presentation</i>	11/2021

Optional

Language: Chinese Mandarin (Native), English (Proficient), Cantonese (Intermediate)

Chartered Financial Analyst (CFA), level I (passed)

GRE: 330/340 (Verbal: 161/170; Math: 169/170) + Writing: 4.0/6.0

TOEFL: 108 (Reading: 29/30; Listening: 28/30; Speaking: 24/30; Writing: 27/30)